

# Macroeconomics

## Expectations, Consumption, and Investment

(Blanchard, Macroeconomics 8th edition Ch. 15)

# Consumption

- ▶ The modern theory of consumption was developed independently in the 1950s
  - ▶ By Milton Friedman as the *permanent* theory of consumption, and
  - ▶ By Franco Modigliani as the *life cycle* theory of consumption.
- ▶ Consumption for a very foresighted consumer depends on:
  - ▶ Financial wealth: The value of checking and saving accounts
  - ▶ Housing wealth: The value of the house owned minus the mortgage due
  - ▶ Human wealth: After-tax labor income over working life
  - ▶ Nonhuman wealth: The sum of financial wealth and housing wealth
- ▶ Total wealth is the sum of human and nonhuman wealth

# Learning from Panel Data Sets

- ▶ Panel data sets are data sets that show the value of one or more variables for many individuals or many firms over time.
- ▶ One example is the Panel Study of Income Dynamics (PSID), which started in 1968 with approximately 4,800 families.
- ▶ Each year, the survey asks people about their income, wage rate, number of hours worked, health, and food consumption.

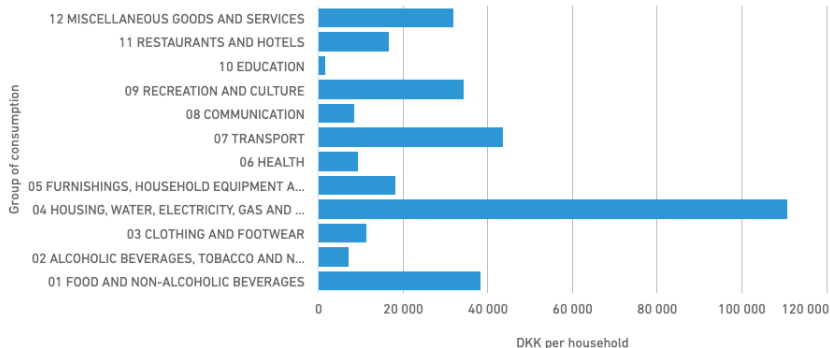
# Denmark Household Budget Survey

- ▶ <https://www.dst.dk/en/Statistik/emner/oekonomi/forbrug/forbrugsundersogelsen>
- ▶ Consumer expectations <https://www.dst.dk/en/Statistik/emner/oekonomi/forbrug/forbrugerforventninger>

## Household average consumption



Price unit: Current prices | Time: 2021:



# Consumption Decision

- ▶ Consumption decision:

$$C_t = C(\text{total wealth}_t)$$

- ▶ Suppose that households want to *smooth* their consumption.
- ▶ If you want to consume the same amount every year, the constant level of consumption that you can afford equals your total wealth divided by your expected remaining life.
- ▶ i.e., spend a proportion of total wealth to maintain roughly the same level of consumption each year throughout life.
- ▶ If that level of consumption were higher than current income, you would borrow the difference.
- ▶ If it were lower than current income, you would save the difference.

## Consumption Decision (con.)

- ▶ This assumes that consumers do think about their wealth and expected future labor income when deciding consumption today.
- ▶ i.e., good (perfect?) foresight
- ▶ However, you might:
  - ▶ not want to plan for constant consumption over your lifetime
  - ▶ make consumption decisions in a less forward-looking fashion
  - ▶ become unemployed or sick
  - ▶ not be able to get a loan from a bank when you need to borrow

# Do People Save Enough for Retirement?

- ▶ Based on a study with the Health and Retirement Study data set, three economists found that retirees have a wealth average of \$1.1 million, enough for a comfortable retirement.
- ▶ Mean Wealth of People, Age 65–69, in 2008 (in 1000s of 2008 dollars):

|                            | Married couples | Single person household |
|----------------------------|-----------------|-------------------------|
| Social Security pension    | 262             | 134                     |
| Employer-provided pension  | 129             | 63                      |
| Personal retirement assets | 182             | 47                      |
| Other financial assets     | 173             | 83                      |
| Home equity                | 340             | 188                     |
| Other equity               | 69              | 18                      |
| Total                      | 1,155           | 533                     |

Source: James M. Poterba, Steven F. Venti, and David A. Wise, "The composition and drawdown of wealth in retirement," *Journal of Economic Perspectives*, 25(4), pages 95–118, Fall 2011.

# Consumption

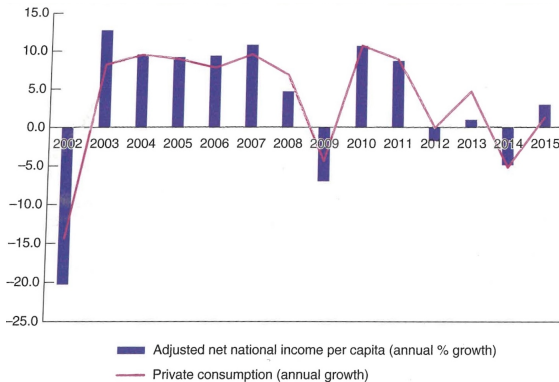
- ▶ Consumption function with after-tax labor income:

$$C_t = C(\overset{\oplus}{\text{total wealth}_t}, \overset{\oplus}{Y_{Lt} - T_t})$$

- ▶ Expectations affect consumption:
  - ▶ Directly through human wealth
  - ▶ Indirectly through nonhuman wealth
- ▶ Implications of expectations of the relation between consumption and income:
  - ▶ Consumption is likely to respond *less than* one-for-one in current income.
  - ▶ Consumption may move even if current income does *not* change.

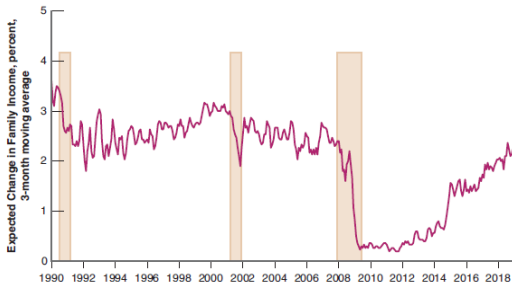
# How Much Do Expectations Matter in Argentina?

- ▶ Argentina: history of booms and busts
- ▶ In 2001-2, Argentina defaulted on its debt, output declined by 20%, and unemployment rate rose to 25%.



## Expected Change in Family Income since 1990

- ▶ Effects of Great Recession on expectations are particularly striking.
- ▶ Relatively stable expectations remained until 2008.
- ▶ After falling sharply in 2008 and 2009, expectations of income growth remained low for a long time, until 2014.



Source: Survey of Consumers, Table 14, University of Michigan,

<https://data.sca.isr.umich.edu>. The shaded areas represent recessions.

## A Two-Period Model

- ▶ Consider a model with two periods: today and tomorrow.
- ▶ Let  $y$  ( $y'$ ) denote income today (tomorrow),  $c$  and  $c'$  denote consumption, and  $s$  denote savings today.
- ▶ The consumer pays lump-sum taxes  $t$  and  $t'$ . The budget constraint today is:

$$c + s = y - t$$

- ▶ If  $s > (<)0$ , then the consumer is a lender (borrower) on the credit market (e.g., bonds with return or real interest rate  $r$ )

# Budget Constraints

- ▶ The budget constraint tomorrow is:

$$c' = y' - t' + (1 + r)s$$

- ▶ Solve to obtain the *lifetime* or *intertemporal* budget constraint:

## Budget Constraints (con.)

# Consumer Preferences

- ▶ We assume that preferences have three properties:
  1. More is always preferred to less.
  2. The consumer likes diversity in his or her consumption bundle: consumption smoothing.
  3. Current consumption and future consumption are normal goods.
- ▶ Recall: demand increases when income increases for normal goods (as opposed to inferior goods).
- ▶ This implies that if there is a parallel shift to the right in the consumer's budget constraint, then current consumption and future consumption both increase.

# Consumer Preferences: Examples

- ▶ Popular utility functions include:
- ▶ Constant relative risk aversion (CRRA) utility function

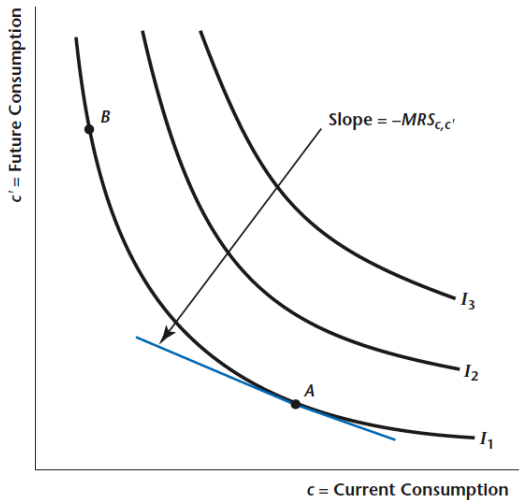
$$u(c) = \frac{c^{1-\eta}}{1-\eta}$$

- ▶ or when  $\eta \rightarrow 1$ , this becomes log utility

$$u(c) = \log(c)$$

- ▶ Do these satisfy the three properties?
- ▶ Represent preferences with indifference map (i.e., family of *indifference curves*)

# Indifference Curves



# Marginal Rate of Substitution (MRS)

- ▶ Denote the **marginal rate of substitution (MRS)** of consumption today for consumption tomorrow as  $MRS_{c,c'}$
- ▶ It is equal to minus the slope of an indifference curve (or the slope of the tangent)
- ▶ A preference for diversity, i.e., diminishing MRS, is captured by the convexity of the indifference curve, which also represents a consumer's desire to smooth consumption over time.
- ▶ If consumption today is large relative to tomorrow, the consumer needs to be given a large quantity of current consumption to willingly give up a small quantity of future consumption  $\Rightarrow$  flatter slope.

## Optimization

- ▶ The consumer maximizes lifetime utility subject to the intertemporal budget constraint, with the discount parameter or weight  $\beta$ :

$$\max u(c) + \beta u(c') \quad \text{s.t.} \quad c + \frac{c'}{1+r} = we$$

- ▶ Substitution gives

$$u(c) + \beta u\left((we - c)(1+r)\right)$$

- ▶ Set the *first-order condition* to zero

$$u'(c) + \beta u'\left((we - c)(1+r)\right)(-(1+r)) = 0$$

$$u'(c) = \beta(1+r)u'\left((we - c)(1+r)\right)$$

$$u'(c) = \beta(1+r)u'(c')$$

# Euler Equation

- ▶ This is known as the **Euler equation** for consumption

$$u'(c) = \beta(1 + r)u'(c')$$

- ▶ It is one of the most famous equations in macroeconomics, lying at the heart of advanced macroeconomic models.
- ▶ It has a beautiful intuition: The consumer must be indifferent between consuming one more unit today on the one hand and saving that unit and consuming in the future on the other.
- ▶ Why? Suppose it does *not* hold, e.g.,  $u'(c) > \beta(1 + r)u'(c')$ , then what?

## MRS Aagain

- ▶ Recall  $MRS_{c,c'}$  gives the slope of an indifference curve, where the  $x$ -axis is consumption today
- ▶ This means that

$$MRS_{c,c'} = \frac{u'(c)}{\beta u'(c')}$$

- ▶ Note the units are  $\frac{\text{utils}}{c \text{ today}} / \frac{\text{utils}}{c' \text{ tomorrow}} = \frac{c' \text{ tomorrow}}{c \text{ today}}$
- ▶ We can rearrange the Euler equation as

$$MRS_{c,c'} = \frac{u'(c)}{\beta u'(c')} = 1 + r$$

# Consumer That Lends

## Log Utility

- ▶ With log utility,  $u(c) = \log(c)$ , which means

$$MRS_{c,c'} = \frac{u'(c)}{\beta u'(c')} = \frac{c'}{\beta c}$$

- ▶ The Euler equation is

$$\frac{c'}{c} = \beta(1 + r)$$

- ▶ The consumer chooses a consumption pattern such that the growth rate of consumption is equal to the product of the discount parameter and the interest rate on savings.
- ▶ The less weight placed on future utility (lower  $\beta$ ), the lower is consumption growth.
- ▶ The higher is the interest rate, the faster is consumption growth.

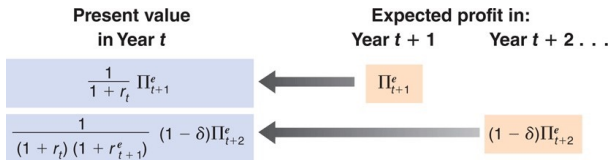
# Increase in Income Today

# Permanent Increase in Income

# Investment

- ▶ To compute the present value of expected profits the firm must first estimate how long the machine will last or the depreciation rate  $\delta$ .
- ▶ The present value of expected profits from buying the machine in year  $t$  is:

$$V(\Pi_t^e) = \frac{1}{1 + r_t} \Pi_{t+1}^e + \frac{1}{(1 + r_t)(1 + r_{t+1}^e)} (1 - \delta) \Pi_{t+2}^e + \dots$$



# The Investment Decision

- ▶ Normalize the price of the machine to 1.
- ▶ Cost-benefit analysis: purchase the machine if  $V(\Pi_t^e) \geq 1$ .
- ▶ The investment function (at the aggregate level) becomes:

$$\oplus$$
$$I_t = I[V(\Pi_t^e)]$$

- ▶ where  $\Pi_t^e$  is the expected profit per unit of capital.
- ▶ James Tobin argued that there should be a tight relation between investment and the value of the stock market.

## Investment

- ▶ Assume static expectations:

$$\begin{aligned}\Pi_{t+1}^e &= \Pi_{t+2}^e = \dots = \Pi_t \\ r_{t+1}^e &= r_{t+2}^e = \dots = r_t\end{aligned}$$

- ▶ Then the (expected) value of buying a machine becomes

$$V(\Pi_t) = \frac{\Pi_t}{r_t + \delta}$$

- ▶ Furthermore, substitution gives

$$I_t = I\left(\frac{\Pi_t}{r_t + \delta}\right)$$

- ▶ where  $r_t + \delta$  is rental cost (user cost), or **shadow cost** of capital. A “rental agency” must charge  $r_t + \delta$  to compensate for their opportunity cost (e.g., sell the machine and buy bonds).

## Investment Function

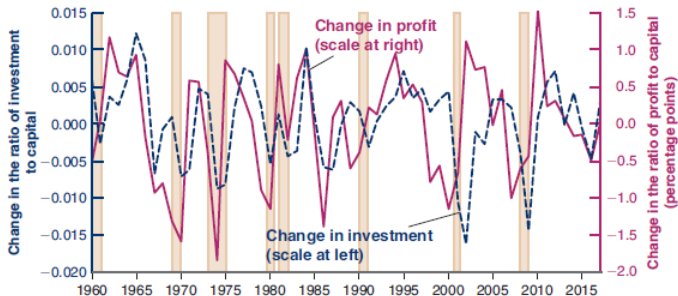
- ▶ Hence, often times we write the profits as  $PY - rK - wN$ .

$$I_t = I\left(\frac{\Pi}{r_t + \delta}\right)$$

- ▶ According to this investment function:
  - ▶ Investment depends on the ratio of profit to the user cost.
  - ▶ The higher the profit, the higher the level of investment.
  - ▶ The higher the user cost, the lower the level of investment.
- ▶ As with consumption, investment may depend on current profits

$$I_t = I[V(\Pi_t^e), \Pi_t]$$

# Changes in Investment and Profit



- *Source:* Gross investment: Federal Reserve Board, Flow of Funds, series FA105013005.A. Capital stock: BEA Fixed Assets Tables, net stock of private nonresidential fixed assets, nonfinancial. Profit: BEA, NIPA Table 1.14, Net operating surplus minus taxes, minus transfers, minus net interest payments.

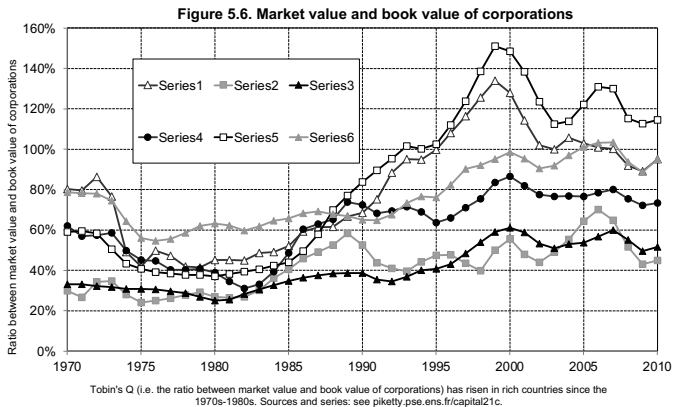
## Investment and the Stock Market

- ▶ Suppose the price per share of a company is \$2, and the purchase price of a machine is \$1. Should the company invest?
- ▶ **Tobin's  $q$**  or  $q$  ratio: value of unit of capital in place relative to current purchase price (i.e., market value to replacement value)

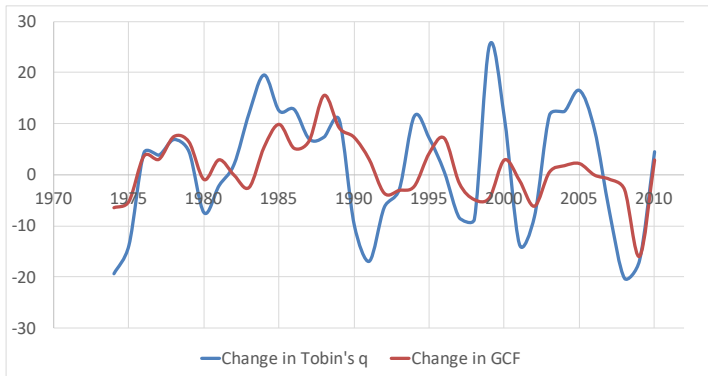
$$q = \frac{\text{Market value of installed capital}}{\text{Replacement cost of installed capital}}$$

- ▶ Mkt value of firm/economy's capital as determined by stock mkt
  - ▶ If  $q > 1$ , managers can raise the market value of firm's stock by buying more capital.
  - ▶ If  $q < 1$ , managers will not replace capital as it wears out.
- ▶ If  $MPK > \text{cost of capital}$ , firms earn profits on installed capital; these firms are more desirable to own, and their market value increases (i.e.,  $q$  increases).
- ▶ As a measure of the incentive to invest,  $q$  reflects the expected future profitability of capital and current profitability.

# Tobin's q Over Time



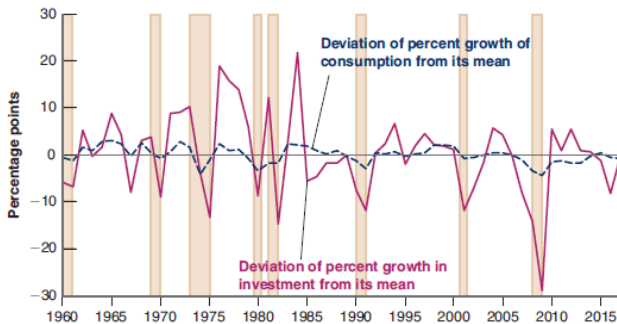
# Tobin's q and Investment



# Volatility of Consumption and Investment

- ▶ Similarities between consumption and investment behavior:
  - ▶ The more transitory consumers expect a current increase in income to be, the less they will increase their consumption.
  - ▶ The more transitory firms expect a current increase in sales to be, the less they revise their assessment of the present value of profits, and thus the less likely they are to buy new machines or build new factories
- ▶ Consumption and investment usually move together.
- ▶ Investment is much more volatile than consumption.
- ▶ Both components contribute roughly equal to output fluctuations.

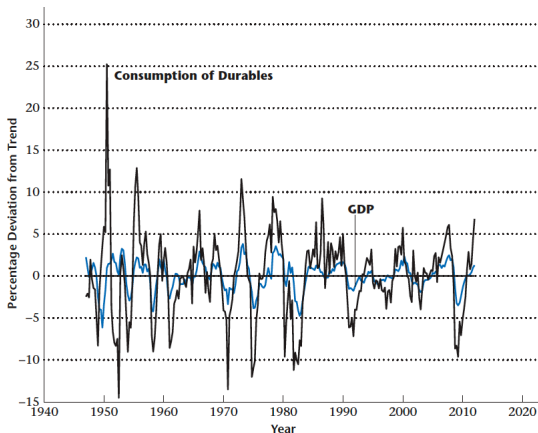
# Rates of Change in Consumption and Investment



- Source: FRED, series PCECC96, GDPDI

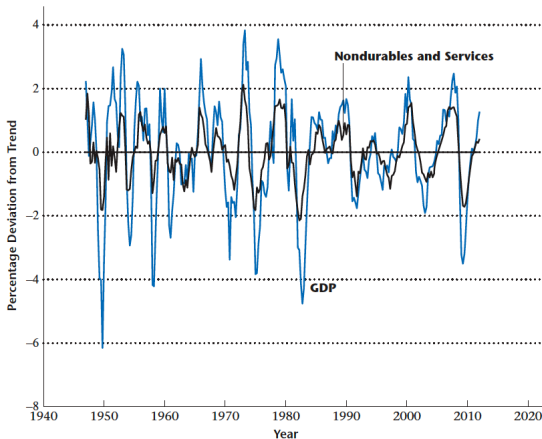
# Durables Consumption

- The consumption of durables is economically similar to investment expenditures, which is why consumer durables expenditure is much more volatile than real GDP.



# Nondurables Consumption

- The consumption of nondurables and services is much smoother than real GDP, reflecting the motive of consumers to smooth consumption relative to income.



# Stock Prices and Consumption

- ▶ Percentage deviations from trend in stock prices and consumption are positively correlated, though stock prices are much more volatile than consumption.

